MARITIME COMMAND QUALIFICATION STANDARD AND PLAN

Mar Eng Tech 00122

Mar Eng Art 00123

Marine Diesel Inspection Course

[Qualification Level]

[Development Period]

Competency Code: 118880

Chapters 1, 2: DD MMM YY
Chapters 3, 4: DD MMM YY

RECORD OF AMENDMENTS

The "Authority" reference indicates approval of the corresponding chapters of this QSP document. Authority for Chapters 1 and 2 of this QSP resides with Chief of Maritime Staff (CMS), Director Maritime Training and Education (DMTE). The Curriculum Control Authority (CCA) for Chapters 3 and 4 is Canadian Forces Naval Engineering School (CFNES). The approval date, the date that the document is checked back in, should also appear in the bottom right corner of every page (the change being made in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. Details on the change(s) must be amplified in Annex J "Course History".

REVISION	APPROVAL DATE	AUTHORITY
QSPWB, 11 June 2010	14 Oct 2010	Chs 3 & 4: O.S.B. Cdr L. Carosielli, CFNES Cmdt

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FOREWORD

1. This training document is developed under the authority of the Chief Maritime Staff and in accordance with the following references:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education; and

   b. A-PD-055-002/PP-002, Marine Engineering MOSID 00123 (Job Based Specification).

2. This publication is effective on receipt. Performance Objectives (POs) are explained fully in Chapter 2. The Lesson Specifications (LSs)/Enabling Objectives (EOs), Enabling Checks (ECs) and Performance Checks (PCs) which specify the standards that the trainee must achieve to become qualified are listed in Chapter 4.

3. Suggestions for amendments may be initiated by anyone. Amendments must be forwarded through the normal chain of command. Specifically, training establishments are to summarize all requested amendments in writing to the applicable approving authority(ies). QSP revision procedures are outlined in the Naval Training Instructions (NTIs). Final approvals for amendments are as follows:

   c. Chapters 1 and 2 – Chief Maritime Staff, Director Maritime Training and Education (DMTE). Suggested amendments must first have the approval of the Curriculum Control Authority Training Establishment; and

   d. Chapters 3 and 4 – Commandant of Canadian Forces Naval Engineering School (CFNES). Training Establishments that conduct training and are not the CCA must forward their suggested amendments to CFNES for approval.

4. The designated CCA(s) for this training is indicated below:

CCA	PHASE	QUALIFICATION
CFNES		TBD

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY CHIEF MARITIME STAFF/DMTE

AIM

1. The aim of the Marine Diesel Inspection Course is to prepare candidates to perform the duties and tasks outlined in the Marine Engineer - Diesel Inspector Job Description (JD).

2. This QSP is designed to document expected performance outcomes and to assist instructors in preparing and organizing themselves to teach the Marine Diesel Inspection course. This document should be read prior to commencing any preparation or scheduling. Any suggestions for the improvement of this document should be brought to the attention of the Quality Control Section MSE Division of CFNES.

COURSE DURATION CALCULATION

3. Course duration is calculated in accordance with MARCORD 9-30.

Actual Training Days (ATT)	13.3
Training Administrative Time (TAT) (if required)	0.5
Base Administrative Time	1
Training Travel Time (if required)	0
Course Duration (total calendar days)	14.8

RESOURCES

4. The following key resources are required to conduct this training. Ranks listed in the following table are the minimum required.

REQUIRED COURSE STAFFING				
TITLE	**RANK**	**QUALIFICATION**	**QTY**	**REMARKS**
Lead Instructor	PO1	Diesel Inspector (AFPL)	1	Positions to be filled by one representative from each coast.
Instructor	PO2	Diesel Inspector (AFPL)	1	
Assistant	LS	Mar Eng Tech	2	

CRITICAL TRAINING RESOURCES	NO. OF DAYS
Operational Diesel Engines	
Large Test Cell Diesel Training Facility in CFNES	

CRITICAL TRAINING RESOURCES	NO. OF DAYS
Small Test Cell Diesel Training Facility in CFNES	
Classroom with IT assets	
Specialty measuring tools	
Boroscope optical	
Boroscope video-probe	

COURSE CAPACITY

5. Planning for the conduct of training is based on the following guidelines:

   a. Maximum – 10; and

   b. Minimum – 6.

COURSE PREREQUISITES

6. Course candidates must:

   a. Hold a Certificate 2 qualification;

   b. Hold the minimum rank of Master Seaman; and

   c. Be recommended by the coastal CDI.

RELATED DOCUMENTS

7. The following documents are related to this QSP:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education;

   b. A-PD-055-002/PP-002, Marine Engineering MOSID 00123 (Job Based Specification); and

   c. C-03-010-551/AM-001, Marine Diesel Inspection Programme (CFTO)

EVALUATION

8. Program evaluation shall be conducted in accordance with the guidance provided in A-P9-050-000 CFITES Vol. 11(1) – Supplement - Evaluation and Validation Techniques.

VALIDATION

9. The training will be validated every five years in accordance with Naval Training Instructions, based on the task list in Annex B.

QUALIFICATION

10. Successful completion of this course constitutes the primary means of attaining the Marine Diesel Inspection Course (TBD).

PRIOR LEARNING ASSESSMENT

11. Prior Learning Assessment shall be conducted in accordance with A-P9-050-000/PT-012 Canadian Forces Military Equivalencies Program and MARCORD 9-47.

SPECIAL INSTRUCTIONS

12. Upon completion of the formal, in-house portion of the Diesel Inspection course, the candidate shall complete the OJPR (Annex E) within a 36 month period. The QSP Writing Board (QSPWB) recommended this OJT period due to possible deployments and other employment demands of shipboard personnel to complete this training.

13. Nominations for DI course are to be sent to Coastal CDI for pre-screening.

14. Instruction to be provided by someone with two years experience as a DI.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY CHIEF MARITIME STAFF/DMTE

PO 001 Carry Out a Diesel Inspection

1. Performance: Carry Out a Diesel Inspection.

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Technical publications;

      (3) Tools and test equipment;

      (4) Assistance;

      (5) Personal Protective Equipment (PPE); and

      (6) Replacement parts.

   b. Denied: N/A

   c. Environmental: At sea/alongside.

3. Standard: The Diesel Inspector shall carry out a diesel inspection (BT0294), IAW refs, by:

   a. Conducting documentation inspection (QSPWB0001), by:

      (1) Collecting documentation (QSPWB0008);

      (2) Analysing documentation (QSPWB0009); and

      (3) Evaluating documentation (QSPWB0006).

   b. Conducting Pre-Inspection Operational Trial (PIOT) (QSPWB0002), by:

      (1) Preparing for trials (DT0501);

      (2) Conducting on-site trials (DT0504); and

      (3) Evaluating PIOT (QSPWB0007).

   c. Conducting external inspection (QSPWB0003), by:

      (1) Inspecting fluid systems (QSPWB0010);

      (2) Inspecting air intake and exhaust systems (QSPWB0011);

      (3) Inspecting driven equipment (QSPWB0012);

      (4) Inspecting enclosure systems (FT0842);

      (5) Inspecting control and monitoring systems (QSPWB0013); and

(6) Evaluating external inspection (QSPWB0014).

d. Conducting internal inspection (QSPWB0004), by:

(1) Verifying engine tolerances (FT0827);

(2) Conducting internal video-probe / boroscope inspection (QSPWB0015);

(3) Verifying fuel delivery timing (FT0821); and

(4) Evaluating internal inspection (QSPWB0016).

e. Conducting operational performance tests (QSPWB0005), by:

(1) Preparing for trials (DT0501);

(2) Conducting on-site trials (DT0504); and

(3) Evaluating operational performance tests (QSPWB0017).

f. Completing diesel engine inspection report, by

(1) Drafting a diesel inspection report (BT0295); and

(2) Presenting diesel inspection reports.

4. References: A1, A2, A3, A6, A10, A11, A12, A13, A17, A18, A20, A21, A22, A24, A25, A26, A27, A29, A30, A31, A33, A34, A35, A37, A38, A39, A40, A41, A42, B1, B2, C1, C2, C4, and C6.

5. Specification Task Numbers: BT0222, BT0240, BT0243, BT0262, BT0293, BT0294, BT0295, CT0424, DT0501, DT0504, DT0508, DT0509, DT0513, DT0514, DT0515, DT0518, DT0539, DT0566, FT0820, FT0821, FT0823, FT0826, FT0827, FT0842, FT0896, FT0901, FT0906, FT0934, FT0939, FT0965, FT1010, FT1011, FT1016, FT1021, FT1030, FT1031, FT1033, FT1046, FT1069, FT1070, HT1301, HT1389, HT1417, HT1432, HT1497, HT1505, JT1730, KT1859, LT1956, MT2183, MT2184, MT2185, QSPWB0001, QSPWB0002, QSPWB0003, QSPWB0004, QSPWB0005, QSPWB0006, QSPWB0007, QSPWB0008, QSPWB0009, QSPWB0010, QSPWB0011, QSPWB0012, QSPWB0013, QSPWB0014, QSPWB0015, QSPWB0016, QSPWB0017.

6. Supporting Skills: N/A

7. Supporting Knowledge: N/A

8. Training Limitations: Training shall be conducted at CFNES Diesel Training Facility (DTF) due to the availability of operational trainers; OJT shall be conducted by DI cells on both coasts.

PO 002 Troubleshoot a Diesel Engine

1. Performance: Troubleshoot a Diesel Engine.

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Technical publications;

      (3) Tools and test equipment;

      (4) Assistance;

      (5) PPE; and

      (6) Replacement parts.

   b. Denied: N/A

   c. Environmental: At sea/alongside.

3. Standard: The Diesel Inspector shall troubleshoot a diesel engine (FT0819), IAW refs, by:

   a. Stating the problem:

   b. Listing all possible causes;

   c. Developing a detailed list of all possible causes;

   d. Developing a theory; and

   e. Providing a course of corrective action.

4. References: A1, A26, A31, A37, A40, A41, B1, B2, C5

5. Specification Task Numbers: BT0240, DT0508, ET0668, FT0814, FT0819.

6. Supporting Skills: N/A

7. Supporting Knowledge: N/A

8. Training Limitations: Training shall be conducted at CFNES DTF due to the availability of operational trainers.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT / COMMANDING OFFICER OF CANADIAN FORCES NAVAL ENGINEERING SCHOOL

GENERAL

1. This is a pass/fail course. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and must be successfully completed to achieve this qualification.

PERFORMANCE CHECKS

2. Summative evaluation for this course is in the form of PCs and shall be conducted at the termination of training for each PO. See Annex K Evaluation Matrix to determine type and resource requirements for all summative and formative assessment. As per MARCORDS 9-60, PCs are tests to assess the accomplishment of the performance under the conditions and to the standards stated in the PO. The following indicates pass marks for all Summative assessment:

   a. All practical PCs will have a pass mark of 60%.

3. Instructional staff may offer students remedial assistance IAW CFNES SO 415 – CFNES Training Framework promulgated by the Commandant, CFNES Halifax.

4. Supplemental PCs must be in the same format and have the same standard as the original test.

5. Formative assessment, in the form of ECs and assignments are not critical unless otherwise indicated in the EO test details. Formative assessment enables student feedback on the assessment results, which is essential for development and learning. Results of ECs are not to be counted towards the students' final grade.

RESPONSIBILITY FOR TESTING

6. Quality control staff provides a measure of quality control by confirming that POs are achieved according to the standards designated in the QSP. They shall be responsible for the design, administration and scoring of all PCs where possible. QC staff must monitor all PC and critical EC rewrites. The administration and scoring of ECs will be the responsibility of the Instructional Staff.

PROGRESS MONITORING

7. Student Progress monitoring including unsatisfactory progress shall be conducted IAW CFNES SO 415.

UNSATISFACTORY TRAINING PROGRESS

8. The training review process is contained in MARCORD 9-60 Volume 1.

TRAINING REVIEW BOARD

9. Training Review Board requirements are contained in MARCORD 9-60 Volume 1 and IAW CFNES SO 425.

COURSE REPORTS

10. Course reports (CF 377) will be prepared IAW DAOD 5031-9 and NTI 01/2008. The narrative in block 17 shall indicate the trainee's strengths and areas for improvement in relation to all POs and provide information of use to the trainee's Commanding Officer and Career Manager. Training limitations or incomplete training must also be identified on the course report.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT / COMMANDING OFFICER OF CANADIAN FORCES NAVAL ENGINEERING SCHOOL

PO 001 Carry Out a Diesel Inspection

EO 001.01 Conduct Documentation Inspection

1. Performance: Conduct documentation inspection.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) Assistance;

      (3) MWM 602 DG log books;

      (4) GM 149 DG log books;

      (5) MWM 226 DDFP log books;

      (6) Past inspection reports;

      (7) Marine Diesel Engine Logs (MDEL); and

      (8) Applicable technical manuals.

   b. Denied: N/A

   c. Environmental: Classroom

3. Standard: The trainee shall conduct documentation inspection (QSPWB0001), IAW refs, by:

   a. Collecting the following documentation;

      (1) Marine Diesel Inspection Programme (DIP);

      (2) Engine specific CFTO's;

      (3) Diesel Inspector Handbooks;

      (4) Log readings; and

      (5) EHM records.

   b. Analysing documentation, specifically;

      (1) Log readings;

      (2) EHM records; and

      (3) Ship's holdings.

c. Evaluating documentation, by;

(1) Identifying outstanding defects;

(2) Documenting deviations from SOP's; and

(3) Grading documentation inspection.

4. Time: Total 7.3 x 50 min periods

a. Lecture 7.3 periods.

5. Method / Media:

a. Lecture;

b. Instructor to facilitate interactive research method with provided manuals;

c. The coastal CDI will present a guest lecture to provide and overview of the DIP; and

d. OJT conducted at coastal DI cell.

6. References:

a. A1 - C-03-010-551/AM-001 - Marine Diesel Inspection Programme;

b. A11 – C-03-010-547/MS-001 - The Oil and Coolant Condition Analysis Program (OCCAP);

c. A12 – C-24-010-020/TB-001 - Corrosion Control Procedure Diesel Engine Cooling Water Treatment;

d. A13 – C-24-020-001/TB-001 - Testing of Marine Diesel Engine Lubricating Oil;

e. A17 – C-27-A08-000/NY-001 - Planned Maintenance Schedule, Engine;

f. A18 – C-27-A08-000/NY-Z01 - Planned Maintenance Schedule, Engine;

g. A21 – C-24-548-000/NY-002 - Preventative Maintenance Schedule. Pielstick 20 PA6 V 280;

h. A22 – C-24-548-000/TB-001 - Technical Bulletin. Pielstick 20 PA6 V 280;

i. A24 – C-97-346-000/NY-002 - Enclosure Fine Water Spray Planned Maintenance Schedule;

j. A25 – C-24-673-000/NY-001 - Instruments and Safety Systems (Pielstick Engine) Local Control Cubicle Automatic Controls;

k. A26 – C-27-A99-000/MS-001 - Operation and Maintenance Manual, Engine;

l. A27 – C-27-A99-000/NY-001 - Planned Maintenance Schedule, Engine;

m. A30 – C-26-411-000/DU-001 - Marine Diesel Equipment Log. 850 kW DG Set;

n. A31 – C-26-411-000/MS-001 - Operation and Maintenance Manual, Engine. 850 kW DG Set;

o. A33 - C-26-411-000/MY-001 - Planned Maintenance Schedule. Engine, 850 kW DG Set;

p. A34 - C-26-411-000/TB-001 - Technical Bulletin MWM TBD602V16K Diesel Engine;

q. A35 - Message DMSS 3376 DTG 101457Z Apr 08 - Kittiwake Diesel Lube Oil Test Kit;

r. A37 - Message DMSS 3382 DTG 051631Z Mar 09 - MWM 602 DG Failures and Revised Operating Instructions;

s. A38 - Fleet Technical Instruction MS 005/08 - Generator Inspections;

t. A40 - C-26-348-000/MS-001 - GM 12V-149 Technical Manual;

u. A41 - C-03-005-033/AA-000 - Naval Engineering Manual (NEMS);

v. A42 - Message DMSS 3 261433Z Nov 09 - Operating Instructions for MWM's Post Bearing Change;

w. B1 - S9086-HB-STM-010 - Naval Ship's Technical Manual for Chapter 233;

x. B2 - S9233-CJ-HBK-020 - US Navy Diesel Engine Inspector Handbook Part II;

y. C1 - Service Bulletin 481251 - Valve Lash Clearance Adjustment;

z. C2 - Service Bulletin 482425 - Timing Gear Backlash Measurement; and

aa. C4 - Dynalco Operating Instructions.

7. Training Aids:

a. DIP handout (ref A1) supplied to each student;

b. Naval Ship's Technical Manual for Chapter 233 (ref B1) supplied to each student;

c. U.S. Navy Diesel Engine Inspector Handbook Part II (ref B2) supplied to each student;

d. Sample past Diesel Inspection Reports will be supplied to each student; and

e. Applicable CFTO's.

8. Test Details:

a. EOs 001.01, 001.02, and 001.03 will be assessed by a written EC at the completion of EO 001.03; and

b. PO 001 will be assessed by a practical PC as detailed on OJPR.

9. Remarks: N/A

10. Teaching Points:

	Description	Method	Time	Reference
TP1	**DIP Overview**			
	• Explain the DIP	Lecture	50 min	A1
	• Explain the Inspection process	Lecture	25 min	A1
TP2	**Collecting the following documentation**			
	(1) DIP;			
	• Ustilize engine specific inspection reports	Lecture OJT	25 min	A1 Annex G
	(2) Engine specific CFTO's;			
	• Technical bulletins	Lecture OJT	10 min	A22, A34
	• Service bulletins	Lecture OJT	10 min	C1, C2
	• Engine specific messages	Lecture OJT	10 min	A37, A38
	(3) Diesel Inspector Handbooks;			
	• Naval Ship's Technical Manual for Chapter 233	Lecture OJT	50 min	B1
	• U.S. Navy Diesel Engine Inspector Handbook Part II	Lecture OJT	50 min	B2
	(4) Log readings			
	• Previous DI report	Lecture OJT	5 min	A1
	(5) EHM Records	Lecture OJT	5 min	A11
TP3	**Analysing documentation, specifically;**			
	(1) Log Readings;			
	• Outstanding deficiencies	Lecture OJT	5 min	A1
	• Oil consumption log	Lecture OJT	5 min	A41

	• Kittiwake log • Viscosity • TBN • Insolubles • Water-in-oil	Lecture OJT	10 min	A13, A35
	• Non-running log	Lecture OJT	5 min	A41
	• De-coking log	Lecture OJT	5 min	A42
	• Running log	Lecture OJT	5 min	A41
	• Associated TB's/FTI's	Lecture OJT	10 min	A22, A34
	• MDEL	Lecture OJT	10 min	A30
	• PM/CM thru CMIS-S/DRMIS	Lecture OJT	10 min	A17, A18, A21, A22, A24, A25, A27, A33
	(2) EHM records;			
	▪ OCCAP	Lecture OJT	10 min	A11
	▪ EDEA data	Lecture OJT	10 min	C4
	(3) Ship's holdings • Spare parts • Applicable tools	Lecture OJT	10 min	A1
TP4	**Evaluating Documentation, by;**			
	(1) Identifying outstanding defects.	Lecture OJT	10 min	A1, A11, A12, A13, A26, A31, A40
	(2) Documenting deviations from SOP's.	Lecture OJT	15 min	A1, A11, A12, A13, A26, A31, A40
	(3) Grading documentation inspection.	Lecture OJT	5 min	A1, A11, A12, A13, A26, A31, A40

PO 001 Carry Out a Diesel Inspection

EO 001.02 Conduct Pre-Inspection Operational Trial

1. Performance: Conduct Pre-Inspection Operational Trial (PIOT).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) Assistance;

      (3) Operational MWM 602 DG;

      (4) Operational GM 149 DG;

      (5) Operational MWM 226 DDFP;

      (6) PPE;

      (7) Electronic Diesel Engine Analyzer (EDEA) test equipment; and

      (8) Special tools; and

      (9) Specified references.

   b. Denied: N/A

   c. Environmental:

      (1) Classroom;

      (2) Small test cell; and

      (3) Large test cell.

3. Standard: The trainee shall conduct PIOT (QSPWB0002), IAW refs, by:

   a. Preparing for trials, by;

      (1) Identifying outstanding defects; and

      (2) Coordinating PIOT.

   b. Conducting on-site trials, by;

      (1) Assisting with EDEA;

      (2) Testing machinery trip gear;

      (3) Conducting functional tests/trials; and

(4) Conducting engine diagnostics.

c. Evaluating PIOT, by;

(1) Identifying outstanding defects;

(2) Documenting deviations from SOPs;

(3) Analysing trial results;

(4) Grading PIOT; and

(5) Determining work requirements.

4. Time: Total 8 x 50 min periods;

a. Lecture 3 periods; and

b. Demo 5 periods.

5. Method / Media:

a. Lecture; and

b. Demonstration of PIOT will be carried out on MWM 602 DG, GM 149 DG, and MWM 226 DDFP by the instructor and assisted by the students.

c. EDEA lecture and demonstration will be delivered by FMFCS Mechanical Engineering EHM personnel.

6. References:

a. A1 - C-03-010-551/AM-001 - Marine Diesel Inspection Programme;

b. A11 – C-03-010-547/MS-001 - The Oil and Coolant Condition Analysis Program (OCCAP);

c. A26 – C-27-A99-000/MS-001 - Operation and Maintenance Manual, Engine;

d. A31 – C-26-411-000/MS-001 - Operation and Maintenance Manual, Engine. 850 kW DG Set;

e. A40 - C-26-348-000/MS-001 - GM 12V-149 Technical Manual;

f. A41 - C-03-005-033/AA-000 - Naval Engineering Manual (NEMS);

g. B2 - S9233-CJ-HBK-020 - US Navy Diesel Engine Inspector Handbook Part II; and

h. C4 - Dynalco Operating Instructions.

7. Training Aids: Applicable technical manuals.

8. Test Details:

a. EOs 001.01, 001.02, and 001.03 will be assessed by a written EC at the completion of EO 001.03; and

b. PO 001 will be assessed by a practical PC as detailed on OJPR.

9. Remarks: N/A

Teaching Points:

	Description	Method	Time	Reference
TP1	**Preparing for trials, by;**			
	(1) Identifying outstanding defects;			
	• Prepare test and trial sheets	Lecture OJT	10 min	A1 Annex G
	• Visual inspection of engines	Demo OJT	50 min	A1 Annex C
	• Visual inspection of support systems	Demo OJT	15 min	A1 Annex C
	(2) Coordinating PIOT, by; • Contacting EDEA personnel • Contacting the Project Leader (man aloft) • Contacting the Ship's staff	Lecture OJT	15 min	A1
TP2	**Conducting on-site trials, by**;			
	(1) Assisting with EDEA;			
	• Introduction	Lecture	10 min	A1, C4
	• Purpose	Lecture	10 min	A1, C4
	• Application	Lecture	10 min	A1, C4
	• Diesel Trap set-up	Lecture	10 min	A1, C4
	• Collecting Data	Lecture	10 min	A1, C4
	• Dump process	Lecture	10 min	A1, C4
	• Input of info and readings	Lecture	10 min	A1, C4
	• Analyze Data	Lecture OJT	10 min	A1, C4
	• Set-up EDEA	Demo	10 min	A1, C4
	(2)Testing machinery trip gear;			
	• Conduct pre-start checks	Demo OJT	15 min	A1, A26, A31, A40

	• Testing fuel stop	Demo OJT	15 min	A1, A26, A31, A40
	• Testing air flaps	Demo OJT	5 min	A1, A31
	(3) Conducting functional tests/trials (engine unloaded);			
	• Flash engine IAW SOP's			
	▪ Test local start/stop	Demo OJT	10 min	A1, A26, A31, A40
	▪ Test overspeed trip	Demo OJT	10 min	A1, A26, A31, A40
	▪ Test lube oil pressure trip	Demo OJT	10 min	A1, A26, A31, A40
	▪ Test remote start/stop	Demo OJT	10 min	A1, A26, A31, A40
	▪ Rounds IAW SOP's	Demo OJT	30 min	A1, A26, A31, A40, A41
	(4) Conducting engine diagnostics (engine under load);			
	• Extract fluid samples	Demo OJT	5 min	A1, A11
	• Take instrumentation readings	Demo OJT	10 min	A1, A26, A31, A40, A41
	• Collect EDEA data	Demo OJT	30 min	A1, C4
	• Process EDEA data	Demo OJT	10 min	A1, C4
	• Analyse exhaust comparison IAW Ringelmann chart	Demo OJT	15 min	B2
TP3	**Evaluating PIOT, by;**			
	(1) Identifying outstanding defects.	Lecture OJT	10 min	A1, A26, A31, A40
	(2) Documenting deviations from SOPs.	Lecture OJT	10 min	A1, A26, A31, A40
	(3) Analysing trial results.	Lecture OJT	15 min	A1, A26, A31, A40
	(4) Grading PIOT.	Lecture OJT	5 min	A1, A26, A31, A40
	(5) Determining work requirements.	Lecture OJT	5 min	A1, A26, A31, A40

PO 001 Carry Out a Diesel Inspection

EO 001.03 Conduct External Inspection

1. Performance: Conduct external inspection.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) Assistance;

      (3) Operational MWM 602 DG;

      (4) Operational GM 149 DG;

      (5) Operational MWM 226 DDFP;

      (6) PPE;

      (7) Special tools; and

      (8) Specified references.

   b. Denied: N/A

   c. Environmental:

      (1) Classroom; and

      (2) Large test cell.

3. Standard: The trainee shall conduct external inspection (QSPWB0003), IAW refs, by;

   a. Inspecting fluid systems, to include;

      (1) Saltwater systems;

      (2) Fresh water systems;

      (3) Fuel systems; and

      (4) Lube oil systems.

   b. Inspecting air intake and exhaust systems;

   c. Inspecting driven equipment;

   d. Inspecting enclosures systems;

e. Inspecting control and monitoring systems; and

f. Evaluating external inspection, by;

(1) Identifying outstanding defects;

(2) Documenting deviations from SOPs;

(3) Grading external inspection; and

(4) Determining work requirements.

4. Time: Total 9 x 50 min periods;

a. Lecture 1 period;

b. Demo 7 periods; and

c. EC 1 period.

5. Method / Media:

a. Lecture;

b. Demo with student assist; and

c. Instructor to demonstrate testing of saltwater thermostat on MWM 602.

6. References:

a. A1 - C-03-010-551/AM-001 - Marine Diesel Inspection Programme;

b. A26 – C-27-A99-000/MS-001 - Operation and Maintenance Manual, Engine;

c. A29 - C-26-321-000/MS-002 - Operation and Maintenance Manual, Enclosure 850 kW DG Set;

d. A31 – C-26-411-000/MS-001 - Operation and Maintenance Manual, Engine. 850 kW DG Set;

e. A40 - C-26-348-000/MS-001 - GM 12V-149 Technical Manual; and

f. A41- C-03-005-033/AA-000 - Naval Engineering Manual (NEMS).

7. Training Aids: Applicable technical manuals.

8. Test Details:

a. EOs 001.01, 001.02, and 001.03 will be assessed by a written EC at the completion of EO 001.03; and

b. PO 001 will be assessed by a practical PC as detailed on OJPR.

9. Remarks: N/A

10. Teaching Points:

	Description	Method	Time	Remarks
TP1	Inspecting external inspection	Lecture	15 min	
TP2	**Inspecting fluid systems, to include;**			
	(1) Saltwater systems;	Demo with Student Assist OJT	30 min	A1, A26, A31, A40, A41
	o Test saltwater thermostat	Demo with Student Assist OJT	50 min	A31, Part 3, Section 2, para 32 – 36, pg. 3-2-39
	(2) Fresh water systems.	Demo with Student Assist OJT	30 min	A1, A26, A31, A40, A41
	(3) Fuel systems.	Demo with Student Assist OJT	30 min	A1, A26, A31, A40, A41
	(4) Lube oil systems.	Demo with Student Assist OJT	30 min	A1, A26, A31, A40, A41
TP3	**Inspecting air intake and exhaust systems.**	Demo with Student Assist OJT	30 min	A1, A26, A31, A40, A41
TP4	**Inspecting driven equipment;**			
	• Generators	Demo with Student Assist	30 min	A1, A26, A31, A40, A41
TP5	**Inspecting enclosures systems;**	OJT		
	• Engine Mounts	Demo with Student Assist OJT	30 min	A1, A26, A29, A31, A40, A41
	• General	Demo with Student Assist OJT	30 min	A1, A26, A29, A31, A40, A41
TP6	**Inspecting control and monitoring systems;**			

	• Engine instrumentation	Demo with Student Assist OJT	30 min	A1, A26, A31, A40, A41
	• Engine shut-down and alarm panel	Demo with Student Assist OJT	30 min	A1, A26, A31, A40, A41
TP7	**Evaluating external inspection, by;**			
	(1) Identifying outstanding defects.	Lecture OJT	10 min	A1, A26, A31, A40
	(2) Documenting deviations from SOP's.	Lecture OJT	10 min	A1, A26, A31, A40
	(3) Grading external inspection.	Lecture OJT	5min	A1- para, 15, page 6 and 7
	(4) Determining work requirements.	Lecture OJT	10min	A1, A26, A31, A40

ENABLING CHECK – EO 001.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.03 will be a written test with multiple choice questions based on content from EO 001.01 through EO 001.03.

3. Pass mark will be 60%.

4. Time required: 1 period.

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.03 will be a written test multiple choice questions based on content from EO 001.01 through EO 001.03.

2. Ensure your desk is clear of material except for the test sheet necessary for this exam.

3. Do not start until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark is 60%

10. You will have 1 x 50 minute period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	YES	NO	COMMENTS

PO 001 Carry Out a Diesel Inspection

EO 001.04 Conduct Internal Inspection

1. Performance: Conduct internal inspection.
2. Conditions:
   a. Given:
      (1) Supervision;
      (2) Assistance;
      (3) MWM 602 DG;
      (4) GM 149 DG;
      (5) MWM 226 DDFP;
      (6) PPE;
      (7) Special Tools; and
         (a) Dial Indicator with Magnetic Base;
         (b) Crankcase Deflection Gauge;
         (c) PDE static display;
         (d) Video-probe; and
         (e) Boroscope.
      (8) Specific References.
   b. Denied: N/A
   c. Environmental:
      (1) Classroom; and
      (2) Large test cell.
3. Standard: The trainee shall conduct internal inspection (QSPWB0004), IAW refs, by;
   a. Verifying engine tolerances, by:
      (1) Measuring camshaft axial thrust;
      (2) Measuring crankshaft axial thrust;
      (3) Measuring crankshaft deflection readings;

(4) Inspecting engine bearings; and

(5) Measuring super-charger/ turbo-charger clearances.

b. Conducting internal video-probe / boroscope inspection by:

(1) Familiarizing with the video-probe;

(2) Inspecting combustion chamber;

(3) Inspecting engine sump;

(4) Inspecting engine gearing; and

(5) Inspecting heat exchangers.

c. Verifying fuel delivery timing, by;

(1) Explaining fuel injection system;

(2) Setting up specialty tools; and

(3) Measuring fuel delivery timing.

d. Evaluating internal inspection, by;

(1) Identifying outstanding defects;

(2) Documenting deviations from SOPs;

(3) Grading internal inspection; and

(4) Determining work requirements.

4. Time: Total 28.8 X 50 min minute periods.

a. Lecture 6 periods;

b. Demo 4 periods; and

c. Practice 18.8 periods.

5. Method / Media:

a. Lecture;

b. Demonstration;

c. Video; and

d. The following practice will be conducted in groups of two students:

(1) Verifying Engine Tolerances - TP1;

(2) Conducting Internal Inspection with Video-probe/ Boroscope - TP2; and

(3) Verifying Fuel Delivery Timing - TP3.

6. References:

   a. A1 - C-03-010-551/AM-001 - Marine Diesel Inspection Programme;

   b. A2 - C-24-548-000/NU-001 - SEMT Pielstick 20 PA6V280 MPC Inspection Report;

   c. A3 - C-26-419-000/NU-001 - Detroit Diesel 16V-149TIB (Model 9163-7306) Inspection Report;

   d. A6 - C-26-411-000/NU-001 - MWM TBD602V16K Inspection Report;

   e. A10 - C-26-364-000/NU-001 - Detroit Diesel 12V149NA (Model 9123-7000) Inspection Report;

   f. A20 - C-24-548-000/MS-001 - Operation and Maintenance Manual. Pielstick 20 PA6 V 280;

   g. A21 - C-24-548-000/NY-002 - Preventative Maintenance Schedule. Pielstick 20 PA6 V 280;

   h. A26 – C-27-A99-000/MS-001 - Operation and Maintenance Manual, Engine;

   i. A31 – C-26-411-000/MS-001 - Operation and Maintenance Manual, Engine. 850 kW DG Set;

   j. A39 - C-26-419-000/MS-001 - 1000Kw Ship's Service Diesel Generator Set and Auxiliary Equipment;

   k. A40 - C-26-348-000/MS-001 - GM 12V-149 Technical Manual;

   l. B1 – S9086-HB-STM-010 - Canadian Marine Diesel Inspector Training;

   m. B2 - S9233-CJ-HBK-020 - US Navy Diesel Engine Inspector Handbook Part II; and

   n. C5 - Everest Video-Probe Operators Manual.

7. Training Aids: Applicable technical manuals.

8. Test Details:

   a. EOs 001.04, 001.05, and 001.06 will be assessed by a written EC at the completion of EO 001.06; and

   b. PO 001 will be assessed by a practical PC as detailed on OJPR.

9. Remarks: N/A.

10. Teaching Points;

	Description	Method	Time	Reference
TP1	**Verifying engine tolerances, by;**			
	(1) Measuring camshaft axial thrust, by;			A1, A3, A40 sect 01-7-2.
	• Explaining diesel engine camshaft tolerances	Lecture	10 min	
	• Setting up dial indicator	Lecture Demo OJT	10 min 20 min	
	• Thrusting camshaft in both directions and record readings.	Practice 5 Groups of 2 Students @ 20 min OJT	100 min	
	(2) Measuring crankshaft axial thrust, by;			A1, A2, A10, A20 Book 1 Chap 2 pg 2-28 and 2-30 Fig 2-15, A21.
	• Explaining diesel crankshaft tolerances	Lecture	10 min	
	• Set up dial indicator	Lecture Demo OJT	10 min 20 min	
	• Thrusting crankshaft in both directions and record readings.	Practice 5 Groups of 2 Students @ 20 min OJT	100 min	
	(3) Measuring crankshaft deflection readings, by;			
	• Explaining crankshaft deflection tolerances	Lecture	20 min	A1 Annex E Para C Sub 6, A2, A21.
	• Explaining deflection measuring gauge o Purpose o Locations on web o Horizontal/vertical deflections	Lecture	10 min 10 min 10 min	B1-para 233-5.24 to 233-5.25.17,B2-pg 9-23 to 9-27.

	• Setup of deflection gauge o Rotating crankshaft to deflection starting point o Recording reading on deflection gauge at the starting point; o Rotating crankshaft and record readings at the five reading locations; and o Reversing crankshaft rotation and verify previous readings.	Lecture Demo Practice 5 Groups of 2 Students @30 min OJT	20 min 20 min 150 min	A20 Book 1 Cha 4 pg 4-29 to 4-44 Fig 4-23, A21, B2 pg 9-24 Fig 9-8
	• Input deflection readings on graph and analyze deflection Graph	Practice OJT	30 min	Example, PDE graph from fleet
	(4) Inspecting engine bearings, by;			
	• Bearing terminology	Lecture	20 min	B2, pages 10-2 to 10-10
	• Recognizing bearing abnormalities o Wear patterns o Normal wear o Abnormal wear/ overheat o Pass around used bearings	Lecture Demo OJT	 10 min 10 min 20 min 10 min	Bearing handouts, B2 pages 10-1 to 10-16 and 10-24 to 10-38.
	(5) Measuring super-charger/ turbo-charger clearances, by;			
	• Set Up for measuring super-charger lobes	Lecture Demo OJT	10 min 40 min	B2 Chap 18.
	• Measure and record	Practice 5 Groups of 2 Students @20 min OJT	100 min	
	• Set Up for measuring turbo-charger rotor axial and radial movement	Lecture Demo OJT	10 min 15 min	A31 Pg 4-14-4 Fig 4-15-1, B2 Chap 18.
	• Measure and record	Practice 5 Groups of 2 Students @20 min Using two turbochargers concurrently OJT	60 min	

TP2	**Conducting internal video-probe/boroscope inspection, by;**			A1, A26, A31 and A40.
	(1) Familiarizing with the video-probe, by;			C5.
	• Explaining the operation of video-probe	Lecture Demo	10 min 20 min	
	• Explaining proper storage of video-probe	Lecture Demo	5 min 5 min	
	• Engine setup for boroscoping	Lecture	10 min	
	(2) Inspecting combustion chamber.			
	• Cylinder liners, • Pistons, • Intake / exhaust valves	Practice 5 Groups of 2 Students @15 min OJT	75 min	
	(3) Inspecting engine sump.			
	• Connecting rod • Crankshaft bearing	Practice 5 Groups of 2 Students @15 min OJT	75 min	
	(4) Inspecting engine gearing.			
	• Main gearing, • Auxiliary gearing	Practice 5 Groups of 2 Students @15 min OJT	75 min	
	(5) Inspecting heat exchangers.			
	• Lube oil cooler • Enclosure cooler • Fresh water cooler	Practice 5 Groups of 2 Students @15 min OJT	75 min	
TP3	**Verifying fuel delivery timing, by;**			A1, A26, A31 and A40.
	(1) Explaining fuel injection systems.	Lecture	50 min	B2 Chap 15-3, A20, A31, A40 Section14.
	(2) Setting up specialty tools.	Demo OJT	50 min	A31 Part 4 Sec 9 Pages 4-9-12 to 4-9-25.

	(3) Measure fuel delivery timing.	Practice 5 Groups of 2 Students @20 min OJT	100 min	
TP4	**Evaluating internal inspection, by:**			
	(1) Identifying outstanding defects.	Lecture OJT	10 min	A1 Annex E
	(2) Documenting deviations from SOPs.	Lecture OJT	10 min	A1 Annex E
	(3) Grading internal inspection	Lecture OJT	5 min	A1 Annex E
	(4) Determining work requirements.	Lecture OJT	10 min	A1 Annex E

PO 001 Carry Out a Diesel Inspection

EO 001.05 Conduct Operational Performance Test

1. Performance: Conduct operational performance test.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) Assistance;

      (3) Operational MWM 602 DG;

      (4) Operational GM 149 DG;

      (5) Operational MWM 226 DDFP;

      (6) PPE;

      (7) EDEA test equipment; and

      (8) Special tools; and

      (9) Specified references.

   b. Denied: N/A

   c. Environmental:

      (1) Classroom;

      (2) Small test cell; and

      (3) Large test cell.

3. Standard: The trainee shall conduct operational performance test (QSPWB0005), IAW refs, by:

   a. Preparing for trials, by;

      (1) Coordinating PIOT.

   b. Conducting on-site trials, by;

      (1) Assisting with EDEA;

      (2) Testing machinery trip gear;

      (3) Conducting functional tests/trials; and

(4) Conducting engine diagnostics.

c. Evaluating operational performance test, by;

(1) Documenting deviations from SOPs;

(2) Analysing trial results;

(3) Grading operational performance test; and

(4) Determining work requirements.

4. Time: Total time 4 x 50 min periods;

a. Lecture 1 period; and

b. Demo 3 periods.

5. Method / Media:

a. EDEA lecture and demonstration will be delivered by FMFCS Mechanical Engineering EHM personnel;

b. Demonstrations shall not be carried out concurrently (each engine 50 minutes); and,

c. Demonstration of the operational performance test will be carried out by the instructor assisted by the students.

6. References:

a. A1 - C-03-010-551/AM-001 - Marine Diesel Inspection Programme;

b. A11 – C-03-010-547/MS-001 - The Oil and Coolant Condition Analysis Program (OCCAP);

c. A26 – C-27-A99-000/MS-001 - Operation and Maintenance Manual, Engine;

d. A31 – C-26-411-000/MS-001 - Operation and Maintenance Manual, Engine. 850 kW DG Set;

e. A40 - C-26-348-000/MS-001 - GM 12V-149 Technical Manual;

f. A41 - C-03-005-033/AA-000 - Naval Engineering Manual (NEMS);

g. B1 - S9086-HB-STM-010 - Canadian Marine Diesel Inspector Training; and

h. C4 - Dynalco Operating Instructions.

7. Training Aids: Applicable technical manuals.

8. Test Details:

a. EOs 001.04, 001.05, and 001.06 will be assessed by a written EC at the completion of EO 001.06; and

b. PO 001 will be assessed by a practical PC as detailed on OJPR.

9. Remarks: N/A

10. Teaching Points:

	Description	Method	Time	Reference
TP1	**Preparing for trials, by;**			
	(1) Coordinating PIOT, by; • Contacting EDEA personnel • Contacting the Project Leader (man aloft) • Contacting the Ship's staff	Lecture OJT	5 min	A1
TP2	**Conducting on-site trials, by**;			
	(1) Assisting with EDEA;			
	• Collecting Data	Lecture	10 min	A1, C4
	• Analyze Data	Lecture OJT	10 min	A1, C4
	(2)Testing machinery trip gear;			
	• Conduct pre-start checks	Demo OJT	10 min	A1, A26, A31, A40
	• Testing fuel stop	Demo OJT	10 min	A1, A26, A31, A40
	• Testing air flaps	Demo OJT	5 min	A1, A31
	(3) Conducting functional tests/trials (engine unloaded);			
	• Flash engine IAW SOP's			
	▪ Test local start/stop	Demo OJT	5 min	A1, A26, A31, A40
	▪ Test overspeed trip	Demo OJT	5 min	A1, A26, A31, A40
	▪ Test lube oil pressure trip	Demo OJT	10 min	A1, A26, A31, A40
	▪ Test remote start/stop	Demo OJT	10 min	A1, A26, A31, A40
	▪ Rounds IAW SOP's	Demo OJT	30 min	A1, A26, A31, A40, A41

	(4) Conducting engine diagnostics (engine under load);			
	• Take instrumentation readings	Demo OJT	10 min	A1, A26, A31, A40, A41
	• Collect EDEA data	Demo	30 min	A1, C4
	• Process EDEA data	Demo	10 min	A1, C4
	• Analyse exhaust comparison IAW Ringelmann chart	Demo OJT	15 min	B2
TP3	**Evaluating operational performance test, by;**			
	(1) Documenting deviations from SOPs	Lecture OJT	5 min	A1, A11, A26, A31, A40, A41, C4
	(2) Analysing trial results • Explain peak firing pressures	Lecture	5 min	A1, A26, A31, A40, B1-Ch 1
	• Data recorded during Phase 2 is to be compared with Phase 5 and noted in the report.	Lecture OJT	5 min	A1, A26, A31, A40
	(3) Grading operational performance test	Lecture OJT	5 min	A1, A26, A31, A40
	(4) Determining work requirements.	Lecture OJT	5 min	A1, A26, A31, A40

PO 001 Carry Out a Diesel Inspection

EO 001.06 Complete Diesel Inspection Report

1. Performance: Complete diesel inspection report.

2. Conditions:

   a. Given:

      (1) IT assets;

      (2) Access to records;

      (3) Specified references; and

      (4) Assistance.

   b. Denied: N/A

   c. Environmental: Classroom

3. Standard: The trainee shall complete diesel inspection report, IAW refs, by:

   a. Drafting a diesel inspection report, by:

      (1) Carrying out post inspection procedures;

      (2) Analyzing full power data; and

      (3) Evaluating trial results.

   b. Presenting diesel inspection reports, specifically;

      (1) MWM 602 DG report;

      (2) GM 149 DG report; and

      (3) MWM 226 DDFP report.

4. Time: Total 8.6 x 50 min periods;

   a. Lecture 0.2 period;

   b. Practice 2.6 periods;

   c. Guided discussion 1.8 periods;

   d. EC 1 period;

   e. PC 2 periods; and

   f. PC review 1 period.

5. Method / Media:

   a. Lecture;

   b. Practice; and

   c. Each group of students will lead a guided discussion for their respective diesel inspection report to the class.

6. References:

   a. A1 - C-03-010-551/AM-001 - Marine Diesel Inspection Programme;

   b. A11 – C-03-010-547/MS-001 - The Oil and Coolant Condition Analysis Program (OCCAP);

   c. A13 - C-24-020-001/TB-001 - Testing of Marine Diesel Engine Lubricating Oil;

   d. A26 – C-27-A99-000/MS-001 - Operation and Maintenance Manual, Engine;

   e. A31 – C-26-411-000/MS-001 - Operation and Maintenance Manual, Engine. 850 kW DG Set;

   f. A40 - C-26-348-000/MS-001 - GM 12V-149 Technical Manual;

   g. A41 - C-03-005-033/AA-000 - Naval Engineering Manual (NEMS); and

   h. C4 - Dynalco Operating Instructions.

7. Training Aids:

   a. IT assets;

   b. Past reports; and

   c. Applicable references.

8. Test Details:

   a. EOs 001.04, 001.05, and 001.06 will be assessed by a written EC at the completion of EO 001.06; and

   b. PO 001 will be assessed by a practical PC as detailed on OJPR.

9. Remarks: Each group of students will draft a diesel inspection report.

10. Teaching Points:

	Description	Method	Time	Reference
TP1	**Drafting a diesel inspection report, by;**			
	(1) Carrying out post inspection procedures;	Lecture OJT	10 min	A1

	• Brief Ship Staff of engine availability			
	• Muster speciality tools/equipment			
	• Complete job notifications			
	(2) Analyzing full power data;			
	• Compare parameters to OEM Specifications	Practice OJT	10 min	A1, A11, A26, A31, A40, A41, C4
	• EDEA Data	Practice OJT	10 min	A1, A31, C4
	• OCCAP	Practice OJT	10 min	A11, A13
	(3) Evaluating trial results;			
	• Draft report	Practice OJT	100 min	A1, A26, A31, A40,
TP2	**Presenting diesel inspection reports, specifically;**			
	(1) MWM 602 DG report	Guided discussion OJT	30 min	A1, A31
	(2) GM 149 DG report	Guided discussion OJT	30 min	A1, A40
	(3) MWM 226 DDFP report	Guided discussion OJT	30 min	A1, A26

ENABLING CHECK – EO 001.06

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.06 will be a written test based on content from EO 001.04 through EO 001.06.

3. Pass mark will be 60%

4. Time required: 1 x 50 minute period

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.06 will be a written test based on content from EO 001.04 through EO 001.06.

2. Ensure your desk is clear of material except for the test sheet necessary for this exam.

3. Do not start until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark is 60%

10. You will have 1 x 50 minute period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	YES	NO	COMMENTS

PERFORMANCE CHECK – PO 001

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. PC 001 will be a written test based on content from PO 001

3. Pass mark will be 60%

4. Time required: 2 x 50 minute periods

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a written test based on content from PO 001

2. Ensure your desk is clear of material except for the test sheet necessary for this exam.

3. Do not start until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark is 60%

10. You will have 2 x 50 minute periods to complete this exam.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	YES	NO	COMMENTS

PO 002 Troubleshoot a Diesel Engine

EO 002.01 Troubleshoot a Diesel Engine

1. Performance: Troubleshoot a diesel engine.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) Assistance;

      (3) Operational MWM 602 DG;

      (4) Operational GM 149 DG;

      (5) Operational MWM 226 DDFP;

      (6) PPE;

      (7) Special tools; and

      (8) Specified references.

   b. Denied: N/A.

   c. Environmental:

      (1) Classroom;

      (2) Small test cell; and

      (3) Large test cell.

3. Standard: The trainee shall troubleshoot a diesel engine, IAW refs, by;

   a. Stating the problem;

   b. Listing all possible causes;

   c. Developing a detailed list of all possible causes;

   d. Developing a theory; and

   e. Providing a course of corrective action.

4. Time: Total time 26.2 x 50 min periods;

   a. Lecture 1.7 periods;

   b. Demo 4.5 periods; and

c. PC 20 periods.

5. Method / Media:

a. Lecture; and

b. Demonstration of TP2 will be carried out on MWM 602 DG, GM 149 DG, and MWM 226 DDFP by the instructor and assisted by the students.

c. In TP3, the instructor will explain common engine problems in the fleet using specific examples from the DI cells.

6. References:

a. A1 - C-03-010-551/AM-001 - Marine Diesel Inspection Programme;

b. A26 – C-27-A99-000/MS-001 - Operation and Maintenance Manual, Engine;

c. A31 – C-26-411-000/MS-001 - Operation and Maintenance Manual, Engine. 850 kW DG Set;

d. A37 - Message DMSS 3382 DTG 051631Z Mar 09 - MWM 602 DG Failures and Revised Operating Instructions;

e. A40 - C-26-348-000/MS-001 - GM 12V-149 Technical Manual;

f. A41 - C-03-005-033/AA-000 - Naval Engineering Manual (NEMS);

g. B1 – S9086-HB-STM-010 - Canadian Marine Diesel Inspector Training;

h. B2 - S9233-CJ-HBK-020 - US Navy Diesel Engine Inspector Handbook Part II; and

i. Annex L (QSP)

7. Training Aids: Applicable technical manuals.

8. Test Details:

a. PO 002 will be assessed by a practical PC, troubleshooting the MWM 602 DG, 226 DDFP and the GM 149 diesel engines;

b. This PC will assess students individually on troubleshooting. PC shall consist of trainee diagnosing two engine problem scenarios; and

c. Each student will have 1 x 50 minute period to perform each scenario, for a total time of 20 x 50 minute periods.

9. Remarks:

a. Students shall read references A1, B1 and B2 as a concurrent activity while they wait to do the troubleshooting PC; and

b. Troubleshooting shall be carried out on the MWM 602 and GM 149 DG's and the MWM 226 DDFP.

10. Teaching Points:

	Description	Method	Time	Reference
TP1	Stating the problem; • Precisely state the problem and not the symptoms. • The problem statement must be clearly defined.	Lecture	5 min	ANNEX L (QSP)
	Listing all possible causes; • List major system • List components • List reasons	Lecture	5 min	ANNEX L (QSP)
	Developing a detailed list of all possible causes; • Engine familiarization-design, construction, installation and application. • Add or eliminate possible causes. • Collect applicable information. o History o Logs o Performance Tests o Running Hours o Number of Failures o Problem Discovery Method o What happened this time o Previous Maintenance o Recent Repairs • Corrective measures since failure	Lecture	5 min	ANNEX L (QSP)

	Developing a theory; • Develop a plan of action, simplest to hardest • Begin the process of elimination • Add additional causes if discovered during inspection • Use process of elimination until actual cause and not just the result is found • Re-evaluate selection • Review list to ensure cause is the only logical cause • Ensure selection justifies how problem was caused. Support finding with facts and logical reasoning based on OJE and thorough understanding of diesel basics.	Lecture	5 min	ANNEX L (QSP)
	Providing a course of corrective action; • Give solid recommendations to eliminate the problem	Lecture	5 min	ANNEX L (QSP)
TP2	**Demonstrate specific fault finding techniques on;** • MWM 602 • GM 149 • MWM 226	Demo with student assist	75 min 75 min 75 min	A1, A26, A31, A37, A41, B1, B2
TP3	**Explain common engine problems;** • Engine start failures • Electric start system Pneumatic start system • Unusual or erratic operation • Engine noises • Abnormal pressures • Abnormal temperatures • Correlate cyl exh temps and cyl firing pressures and compression pressures • Abnormal speed • Exhaust smoke • Excessive consumption of lube oil, fuel or water • Contamination of lube oil, fuel or water	Lecture Lecture Lecture Lecture Lecture Lecture Lecture Lecture Lecture Lecture Lecture Lecture	5 min 5 min 5 min 5 min 5 min 5 min 5 min 5 min 5 min 5 min 5 min 5 min	A1, A26, A31, A37, A41, B1, B2, B2, ch 24-1, 24-2, ANNEX L (QSP) B2, 24-3 B2, 24-4 B2, 24-5 B2, 24-6 B2, 24-7 B2, 24-8 B2, 24-9 B2, 24-10 B2, 24-11

PERFORMANCE CHECK – PO 002

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Large test cell

Arrangements and Procedures

2. PC 002 will be a practical test based on content from PO 002

3. Pass mark will be 60%.

4. Each student will be presented with two scenarios.

5. Time required: 1 x 50 minute period per scenario.

6. References will be provided.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical test based on content from PO 002.

2. Student will be presented with two individual practical scenarios, of which the student must utilize the Troubleshooting Procedures instructed during EO 002.01 to satisfactorily provide a solution.

3. Do not start until you are told to do so.

4. Work carefully. Safety is paramount.

6. Be sure to cover all aspects of the troubleshooting procedure.

7. If you have any questions/concerns during the test, ask your instructor.

8. References will be provided as required.

9. Pass mark is 60%

10. You will have 1 x 50 minute period to complete each scenario.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	YES	NO	COMMENTS
Annex L, troubleshooting process checklist	x		
Diagnose diesel engine problems	x		The Senior diesel instructor will provide the checklist to the instructor for the specified scenario.

ANNEX A: MAIN REFERENCES

A1 = CF or other official Canadian government publication

A2 = Second and subsequent references in any category

B = Official Allied Force or Allied government publication

C = Commercial publication

REF CODE	REF NUMBER	PUBLICATION TITLE
A1	C-03-010-551/AM-001	Marine Diesel Inspection Programme
A2	C-24-548-000/NU-001	SEMT Pielstick 20 PA6V280 MPC Inspection Report
A3	C-26-419-000/NU-001	Detroit Diesel 16V-149TIB (Model 9163-7306) Inspection Report
A4	C-27-A08-000/NU-001	Detroit Diesel 12V-92TA (Model 8123-7400) Inspection Report
A5	C-27-A99-000/NU-001	MWM D226-6 Inspection Report
A6	C-26-411-000/NU-001	MWM TBD602V16K Inspection Report
A7	C-26-513-000/NU-001	Valenta 16RPA200SZ Inspection Report
A8	C-27-362-000/NU-001	Caterpillar D398 Inspection Report
A9	C-24-609-000/NU-001	Caterpillar 3406B Inspection Report
A10	C-26-364-000/NU-001	Detroit Diesel 12V149NA (Model 9123-7000) Inspection Report
A11	C-03-010-547/MS-001	The Oil and Coolant Condition Analysis Program (OCCAP)
A12	C-24-010-020/TB-001	Corrosion Control Procedure Diesel Engine Cooling Water Treatment
A13	C-24-020-001/TB-001	Testing of Marine Diesel Engine Lubricating Oil
A14	C-27-A08-000/MF-001	SERIES 92 Service Manual
A15	C-27-A08-000/ML-001	Operation and Maintenance Manual
A16	C-27-A08-000/MY-001	Detroit Diesel V-92 Parts Catalogue
A17	C-27-A08-000/NY-001	Planned Maintenance Schedule, Engine
A18	C-27-A08-000/NY-Z01	Planned Maintenance Schedule, Engine
A19	C-24-617-000/MS-001	Caterpillar's Operation and Maintenance Manual
A20	C-24-548-000/MS-001	Operation and Maintenance Manual. Pielstick 20 PA6 V 280
A21	C-24-548-000/NY-002	Preventative Maintenance Schedule. Pielstick 20 PA6 V 280
A22	C-24-548-000/TB-001	Technical Bulletin. Pielstick 20 PA6 V 280
A23	C-97-346-000/NK-001	Enclosure Fine Water Spray System Test Procedure
A24	C-97-346-000/NY-002	Enclosure Fine Water Spray Planned Maintenance Schedule
A25	C-24-673-000/NY-001	Instruments and Safety Systems (Pielstick Engine) Local Control Cubicle Automatic Controls
A26	C-27-A99-000/MS-001	Operation and Maintenance Manual, Engine

REF CODE	REF NUMBER	PUBLICATION TITLE
A27	C-27-A99-000/NY-001	Planned Maintenance Schedule, Engine
A28	C-26-321-000/MS-001	Operation and Maintenance Manual, Electrical. 850 kW DG Set
A29	C-26-321-000/MS-002	Operation and Maintenance Manual, Enclosure 850 kW DG Set
A30	C-26-411-000/DU-001	Marine Diesel Equipment Log. 850 kW DG Set
A31	C-26-411-000/MS-001	Operation and Maintenance Manual, Engine. 850 kW DG Set
A32	C-26-411-000/MY-001	Parts List Engine, Diesel (850 kW Generator Set)
A33	C-26-411-000/NY-001	Planned Maintenance Schedule. Engine, 850 kW DG Set
A34	C-26-411-000/TB-001	Technical Bulletin MWM TBD602V16K Diesel Engine
A35	Message DMSS 3376 DTG 101457Z Apr 08	Kittiwake Diesel Lube Oil Test Kit
A36	UCR E/3555/ER/2001/0002	Closure of Intake/Exhaust Drain Valves June 2005
A37	Message DMSS 3382 DTG 051631Z Mar 09	MWM 602 DG Failures and Revised Operating Instructions
A38	Fleet Technical Instruction MS 005/08	Generator Inspections
A39	C-26-419-000/MS-001	1000Kw Ship's Service Diesel Generator Set and Auxiliary Equipment
A40	C-26-348-000/MS-001	GM 12V-149 Technical Manual
A41	C-03-005-033/AA-000	Naval Engineering Manual (NEMS)
A42	Message DMSS 3 261433Z Nov 09	Operating Instructions for MWM's Post Bearing Change
B1	S9086-HB-STM-010	Naval Ship's Technical Manual for Chapter 233
B2	S9233-CJ-HBK-020	US Navy Diesel Engine Inspector Handbook Part II
C1	Service Bulletin 481251	Valve Lash Clearance Adjustment
C2	Service Bulletin 482425	Timing Gear Backlash Measurement
C3	Service Bulletin 449243 Rev 21	Tightening Torque Table, Fasteners
C4		Dynalco Operating Instructions

REF CODE	REF NUMBER	PUBLICATION TITLE
C5		Everest Video-Probe Operators Manual

ANNEX B: TASK LIST

1. The following tasks are from the following JDs:

    1. Marine Engineer - Diesel Inspector - Junior;
    2. Marine Engineer - Diesel Inspector;
    3. Marine Engineer - Diesel Inspector - Senior; and
    4. Marine Engineer - Chief Diesel Inspector.

JDs	TASK NO.	TASK ELEMENT	TRG / NO TRG	RATIONALE
1, 2, 3, 4	AT0001	Enforce the application of the Naval Divisional System	NT	NCM GS
1, 2, 3, 4	AT0002	Maintain Divisional notes and Unit Employment Record (UER)	NT	NCM GS
1, 2, 3, 4	AT0003	Draft a Personnel Evaluation Report (PER) or a Personnel Development Review (PDR)	NT	NCM GS
2, 3, 4	AT0004	Review a Personnel Evaluation Report or a Personnel Development Review	NT	NCM GS
2, 3, 4	AT0005	Brief and debrief personnel with regards to Performance Evaluation Report or Personnel Development Review results	NT	NCM GS
4	AT0007	Establish the ranking/selection criteria of personnel for a merit board	NT	NCM GS
2, 3, 4	AT0009	Draft an Establishment Change Proposal (ECP)	NT	OJE
4	AT0012	Establish a training priority	NT	OJE
4	AT0013	Produce a departmental annual training plan	NT	OJE
4	AT0018	Invigilate a Performance Check or an Enabling Check	NT	OJE
4	AT0022	Select personnel for training/coursing	NT	OJE
4	AT0025	Coordinate training facilities usage with contractors	NT	OJE
4	AT0030	Review a course report (CF 377)	NT	OJE
4	AT0032	Perform duties as a member of a Training Review Board (TRB)	NT	NCM GS
4	AT0035	Track collective and individual training	NT	OJE
4	AT0036	Monitor training throughput	NT	OJE
1, 2, 3, 4	AT0037	Liaise with other government agencies	NT	OJE
1, 2, 3, 4	AT0038	Liaise with emergency services agencies	NT	OJE
1, 2, 3, 4	AT0040	Perform as a member of a force protection organization or a base security force	NT	NCM GS
1, 2, 3	AT0058	Perform as a member of the base duty watch organization	NT	NCM GS
1, 2, 3, 4	AT0064	Control usage of hazardous materials	NT	QL5
1, 2, 3, 4	AT0065	Implement security Standard Operating Procedures	NT	NCM GS

1, 2, 3	AT0068	Clean up fuel spills	NT	OJE
4	AT0071	Assess proposed changes to unit Standard Operating Procedures	NT	OJE
2, 3, 4	AT0074	Brief senior officers on ship's readiness	NT	OJE
4	AT0077	Determine effect of additional tasking/duties on operations	NT	OJE
4	AT0080	Determine impact of constraints on operations	NT	OJE
1, 2, 3, 4	AT0082	Ensure standards are maintained	NT	OJE
2, 3, 4	AT0083	Evaluate personnel for the award of operating certificates	NT	OJE
1, 2, 3, 4	AT0084	Evaluate standards/criteria used in the workplace	NT	OJE
1, 2, 3, 4	AT0085	Implement repair and maintenance instructions	NT	OJE
1, 2, 3	AT0087	Prepare equipment for shipment or storage	NT	OJE
4	AT0091	Provide input for Command reports and returns	NT	OJE
2, 3, 4	AT0093	Participate as a member of a certification board	NT	OJE
2, 3, 4	AT0094	Conduct certification pre-boards	NT	OJE
2, 3, 4	AT0095	Participate as a member of a certification pre-board	NT	OJE
1, 2, 3, 4	AT0096	Control resources	NT	OJE
1, 2, 3, 4	AT0098	Develop briefs	NT	NCM GS
1, 2, 3, 4	AT0099	Enforce shop safety	NT	NCM GS
2, 3, 4	AT0100	Ensure staff qualifications are maintained	NT	OJE
4	AT0101	Estimate resource requirements and costs	NT	OJE
2, 3, 4	AT0102	Investigate accidents and safety violations	NT	OJE
2, 3, 4	AT0103	Investigate operating procedural errors	NT	OJE
4	AT0104	Liaise with career managers on personnel matters	NT	OJE
1, 2, 3, 4	AT0105	Liaise with other unit/sections on work-related matters	NT	OJE
4	AT0107	Monitor establishment strength	NT	OJE
4	AT0111	Process training requests	NT	OJE
4	AT0112	Provide advice in matters of supply, administration and training	NT	OJE
4	AT0113	Provide status of communication and cooperation	NT	OJE
4	AT0114	Provide status of training	NT	OJE
1, 2, 3, 4	AT0115	Operate Information Technology (IT) systems and equipment	NT	OJE
2, 3, 4	AT0118	Review/edit correspondence and documents	NT	OJE
4	AT0122	Validate training	NT	OJE
1, 2, 3, 4	AT0125	Write military letters	NT	NCM GS
2, 3, 4	BT0200	Perform post accident investigations	NT	OJE
2, 3, 4	BT0201	Determine recommendations to prevent accidents	NT	OJE

1, 2, 3, 4	BT0202	Inspect Configuration Management Information (CMIS) system	NT	OJE
1, 2, 3, 4	BT0207	Review maintenance schedules	NT	OJE
1, 2, 3, 4	BT0210	Direct a preventive maintenance activity	NT	OJE
4	BT0212	Coordinate a preventive maintenance activity	NT	OJE
1, 2, 3, 4	BT0213	Validate preventive maintenance tasks and schedules	NT	OJE
1, 2, 3, 4	BT0214	Direct a corrective maintenance activity	NT	OJE
1, 2, 3, 4	BT0217	Liaise with refit facility for preventive and corrective maintenance	NT	OJE
1, 2, 3, 4	BT0218	Utilize Material Acquisition & Support Information system (MASIS)	NT	IT
1, 2, 3, 4	BT0222	Evaluate maintenance procedures and techniques	T	EO 001.01
4	BT0229	Schedule work with ship's time available	NT	OJE
4	BT0232	Monitor operational deficiency report	NT	QL6
1, 2, 3, 4	BT0240	Evaluate log readings	T	EO 001.01 EO 002.01
1, 2, 3, 4	BT0243	Determine work requirements	T	EO 001.02, .03, .04, .05.
1, 2, 3, 4	BT0252	Requisition technical drawings/publications	NT	OJE
4	BT0253	Update technical drawings/publications	NT	OJE
1, 2, 3, 4	BT0260	Provide expertise for repair facilities	NT	OJE
1, 2, 3, 4	BT0262	Identify Outstanding Defects	T	EO 001.01, .02, .03, .04,
1, 2, 3, 4	BT0265	Draft Unsatisfactory Condition Reports	NT	QL5
1, 2, 3, 4	BT0266	Review Unsatisfactory Condition Reports	NT	QL5
1, 2, 3, 4	BT0272	Review technical documentation	NT	OJE
1, 2, 3, 4	BT0273	Review technical reports	NT	OJE
1, 2, 3, 4	BT0274	Prepare technical directives	NT	IT
1, 2, 3, 4	BT0276	Review Engineering Changes	NT	IT
4	BT0281	Advise senior staff on technical readiness	NT	OJE
4	BT0285	Provide inputs to Terms of Reference for Technical Investigations (TIs)	NT	OJE
4	BT0286	Perform Technical Investigations (TIs) of systems and equipment	NT	OJE
1, 2, 3, 4	BT0287	Perform field investigations	NT	OJE
1, 2, 3, 4	BT0288	Recommend improvements to marine policies and procedures	NT	OJE
4	BT0290	Investigate job requirements	NT	OJE
1, 2, 3, 4	BT0291	Implement job instructions	NT	OJE
4	BT0293	Coordinate diesel inspections	T	EO 001.02
1, 2, 3, 4	BT0294	Carry out diesel inspections	T	PO 001
1, 2, 3, 4	BT0295	Draft diesel inspection reports	T	EO 001.06
4	BT0296	Maintain project folders	NT	OJE
1, 2	BT0297	Perform Lock-out / Tag-out procedures	NT	IT
1, 2, 3, 4	CT0407	Research manufacturers' specifications	NT	OJE

1, 2, 3, 4	CT0424	Document deviations from contract specification or Standard Operating Procedures	T	EO 001.01, .02, .03, .04, .05.
1, 2, 3, 4	DT0500	Develop trials	NT	OJE
1, 2, 3, 4	DT0501	Prepare for trials	T	EO 001.02
1, 2, 3, 4	DT0502	Schedule equipment and system trials	NT	OJE
1, 2, 3, 4	DT0504	Conduct on-site trials	T	EO 001.02
1, 2, 3, 4	DT0505	Witness inspections/tests/trials	NT	OJE
1, 2, 3	DT0506	Act as trial Conducting Authority	NT	OJE
1, 2, 3, 4	DT0507	Produce trial reports	NT	OJE
1, 2, 3, 4	DT0508	Review test and trial reports	T	EO 001.01 EO 002.01
1, 2, 3, 4	DT0509	Analyse trials results - ~~Evaluate trials results~~	T	EO 001.02, .05, .06.
1, 2, 3	DT0513	Prepare trials documentation	T	PO 001
1, 2, 3, 4	DT0514	Carry out post inspection procedures	T	EO 001.06
1, 2, 3	DT0515	Prepare test sheets and trials sheets	T	EO 001.02
1, 2, 3	DT0518	Perform functional test/trial of systems	T	EO 001.02
1, 2, 3, 4	DT0519	Assist outside agencies while conducting tests and trials	NT	OJE
4	DT0539	Analyze full-power data	T	EO 001.06
1, 2, 3, 4	DT0551	Test pneumatic control system	NT	OJE
1, 2, 3	DT0556	Perform hammer test	NT	IT
1, 2, 3	DT0557	Employ electrical / electronic / digital precision measuring test equipment	Remove	Duplicated in MT2383 to 2385
1, 2, 3	DT0566	Test alarm systems	T	EO 001.03
1, 2, 3, 4	ET0668	Diagnose engine condition	T	PO 002
1, 2, 3	FT0814	Replace engine components and accessories	T	PO 002
1, 2, 3	FT0815	Replace engine enclosure components	NT	IT
1, 2, 3	FT0818	Inspect diesel engines	Remove	Duplicated in BT0294
1, 2, 3	FT0819	Troubleshoot diesel engines	T	PO 002
1, 2, 3	FT0820	Time a diesel	T	EO 001.04
1, 2, 3	FT0821	Verify fuel delivery timing ~~Test and adjust diesel injectors~~	T	EO 001.04
1, 2, 3	FT0823	Adjust governors and components	T	EO 001.03
1, 2, 3	FT0826	Take crankshaft deflection readings	T	EO 001.04
1, 2, 3	FT0827	Verify Engine Tolerances ~~Measure engine clearances~~	T	EO 001.04
1, 2, 3	FT0842	Inspect engine enclosure cooling system	T	EO 001.03
1, 2, 3	FT0896	Inspect suction strainers	T	EO 001.03
1, 2, 3	FT0901	Inspect fuel oil systems	T	EO 001.03
1, 2, 3	FT0906	Replace fuel oil systems components	T	EO 001.03
1, 2, 3	FT0934	Inspect pneumatic assemblies	T	EO 001.03
1, 2, 3	FT0939	Inspect pneumatic valve actuators	T	EO 001.03
1, 2, 3	FT0965	Inspect piping	T	EO 001.03
1, 2, 3	FT1010	Adjust mechanical alignment	T	EO 001.04
1, 2, 3	FT1011	Inspect shafts	T	EO 001.04

1, 2, 3	FT1016	Remove/re-install fluid system components	T	EO 001.03
1, 2, 3	FT1021	Check for system leaks	T	EO 001.03
1, 2, 3	FT1030	Inspect condenser and heat exchanger components	T	EO 001.04
1, 2, 3	FT1031	Inspect control systems	T	EO 001.03
1, 2, 3	FT1033	Inspect mechanical seals	T	EO 001.03
1, 2, 3	FT1046	Remove/re-install manifolds	T	EO 001.03
1, 2, 3	FT1069	Take instrumentation readings	T	EO 001.02, .03, .05
1, 2, 3	FT1070	Test machinery trip gear	T	EO 001.02
1, 2, 3	HT1301	Inspect power generation systems	T	EO 001.03
1, 2, 3	HT1389	Check generators and motors for oil and/or water ingress	T	EO 001.03
1, 2, 3	HT1417	Inspect sacrificial anodes	T	EO 001.04
1, 2, 3	HT1432	Inspect diesel engine electrical components	T	EO 001.03
1, 2, 3	HT1497	Adjust engine component clearances	T	EO 001.03
1, 2, 3	HT1505	Inspect grounding straps	T	EO 001.03
1, 2, 3	HT1537	Test alarms	Remove	Duplicated in DT0566
1, 2, 3	JT1730	Use precision measuring tools	T	EO 001.02, .04
1, 2, 3	JT1732	Use hand tools	NT	IT
1, 2, 3	JT1744	Replace bolts, nuts and studs	NT	IT
1, 2, 3	KT1826	Inspect lagging and insulation	NT	IT
1, 2, 3	KT1828	Remove lagging and insulation	NT	IT
1, 2, 3	KT1859	Inspect vibration/shock mounts	T	EO 001.02, .03
1, 2, 3	LT1956	Inspect fitted Firefighting systems	T	EO 001.02, .03
1, 2, 3	MT2120	Provide Engineering Officer of the Watch (EOOW) with shaft torque readings	NT	OJE
1, 2, 3	MT2183	Operate electrical measuring equipment	T	EO 001.02, .03, .05
1, 2, 3	MT2184	Operate electronic measuring equipment	T	EO 001.04
1, 2, 3	MT2185	Operate digital measuring equipment	T	EO 001.02, .03, .05
1, 2, 3	MT2186	Monitor a Liquid Level Management System (LLMS) console	NT	IT
1, 2, 3	MT2214	Operate diesel powered auxiliary equipment	NT	IT
	QSPWB0001	Conduct Documentation Inspection	T	EO 001.01
	QSPWB0002	Conduct Pre-Inspection Operational Trial	T	EO 001.02
	QSPWB0003	Conduct External Inspection	T	EO 001.03
	QSPWB0004	Conduct Internal Inspection	T	EO 001.04
	QSPWB0005	Conduct Operational Performance test	T	EO 001.05
	QSPWB0006	Evaluate documentation	T	

				EO 001.01
	QSPWB0007	Evaluate PIOT	T	EO 001.02
	QSPWB0008	Collect documentation	T	EO 001.01
	QSPWB0009	Analyse documentation	T	EO 001.01
	QSPWB0010	Inspect fluid systems	T	EO 001.03
	QSPWB0011	Inspect air intake and exhaust systems	T	EO 001.03
	QSPWB0012	Inspect driven equipment	T	EO 001.03
	QSPWB0013	Inspect control and monitoring systems	T	EO 001.03
	QSPWB0014	Evaluate external inspection	T	EO 001.03
	QSPWB0015	Conduct internal video-probe / boroscope inspection	T	EO 001.04
	QSPWB0016	Evaluate internal inspection	T	EO 001.04
	QSPWB0017	Evaluate operational performance tests	T	EO 001.05

ANNEX C: QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates:

25 May 2010 to 11 June 2010

Location:

CFNES, Bldg S-88, Room 402

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Fortin (Chairman)	AG	Lt(N)	00344	CFNES
Rivalin (TDO / Supervisor)	GC	Capt.	00211	CFNES
Prince (TDO)	MMM	Lt	00211	CFNES
Harding (DMTE Rep)	GD	CPO1	00126	DMTE
Hatcher	RR	CPO2	00123	CFNES
Pryor	MR	CPO2	00123	FMFCS
Sheals	SP	CPO2	00123	FMFCB
Bernard	WG	PO1	00123	FMFCS
Walters	AG	PO1	00123	CFNES
Warren	BD	PO1	00123	HMCS Preserver
Green (Secretary)	JAG	PO1	00123	HMCS Preserver

Board Objective:

1. The mandate of the Mar Eng Diesel Inspector QSPWB was to;

    a. Develop a Diesel Inspector QSP utilizing job description reports;
    b. Finalize composition of the QSP;
    c. Review and amend QSP formatting IAW CFITES; and
    d. Establish measurable standards.

Specification Task – No Train Decisions:

2. The tasks are listed and detailed with substantiation in Annex B.

Discussions / Decisions:

3. DMTE provided a general overview of the history of the Diesel Inspection Course and the requirement to produce a QSP which necessitated convening a QSPWB. The board agreed that the development of a Diesel Inspector QSP by Fleet SMEs provided the best opportunity to ensure future Diesel Inspector training meets the requirement of the Canadian Navy and conforms to CFITES standards.

4. The first task of the board was to conduct a documentation review. The following explains the review and subsequent finding which enabled the board to determine the correct course of action to create an effective and efficient Diesel Inspector training program.

5. The board identified the four job descriptions related to Diesel Inspection as the primary documents to extract the performance requirement descriptions to perform as a Diesel Inspector. The main tasks form the JDs were listed in a table and the board concluded that the majority of these tasks were duplicated across the JDs. However, the board also identified discrepancies within the JDs compared to the tasks actually performed in the respective jobs in the Diesel Inspection Cell. There was considerable debate regarding the level of involvement in Technical Investigations (TIs),across the four JDs; the board concluded that the DI-JRs and the DIs are involved in TIs, where as DI-SRs and CDIs participate in and/or conduct TIs. Recommendations from the SME's were made concerning each JD, to reflect the job performance requirements and are reflected in the following documents using track changes:

Jr DI JD DI JD Sr DI JD CDI JD

6. A review of the DI CFTO C-30-010-551/AM-001 Marine Diesel Inspection Programme confirmed that a number of changes have to be made in order to align the document with the four JDs. In the current Marine Diesel Inspection Programme document, references made to the "Snr DI" job are actually aligned to the CDI JD. Currently, the document specifies that to a FMF Diesel Specialist is required for the oral board sign off on the OJPR. The board recommended changing the DI Qualification, para 6, line 13, to read: "The examining board shall include as a minimum the CDI and the candidate's inspection supervisor and chaired by the SSO East Coast/ MESO West Coast and may include the FMF Diesel Specialist". This change will allow some flexibility for the board to assess the student if the FMF Diesel Specialist is not available. The SMEs referred to the Marine Diesel Inspection Programme, as the governing CFTO used by Diesel Inspectors, to create additional tasks required in the QSP to reflect the job performance of the Diesel inspector that are not in the JDs.

7. The board reviewed the Mar Eng Tech QL5 QSP, PO 003, Repair CPF Diesel Engines. The board agreed that the QL5 provided a reference to complete the task analysis and that the changes implemented during the last Mar Eng Tech QL5 QSPWB had a significant impact on which tasks required training for a Diesel Inspector. However, It was observed that there were significant variations between the Mar Eng Tech QL5 PO 003 on Troubleshooting and the process and procedures employed when

troubleshooting as a diesel inspector. As a result of this comparison, the board decided that the process and procedure that had previously been taught on the Diesel Inspector course was representative of how DIs actually troubleshoot on the job and was reviewed and amended to be inserted into the current QSP. Initially, SMEs attempted to insert troubleshooting throughout the inspection phases of PO 001; board members agreed that it was ineffective as troubleshooting and diesel inspection are carried out independently and therefore must be separate within the QSP. DMTE agreed that troubleshooting should be included in the QSP as a separate PO and the SMEs developed a scalar to capture the troubleshooting process.

8. When discussing assessment for PO 002 Troubleshoot Diesel Engines, the board decided that the best assessment instrument would be a PC comprised of 2 scenarios per student of progressive difficulty with faults staged in a diesel engine. The board created Annex L Troubleshooting Process Checklist to be used for assessment of trainees as well as Annex M Troubleshooting Guide to be used by trainees as a training aid.

9. The board concluded that an oral board is required as the final sign-off on the OJPR to validate the candidate's knowledge and understanding of the diesel engine inspection program. The OJPR includes guidance for the composition of the oral board as well as the type and number of questions to be asked. Previously, qualification boards had been hosted at CFNES MSE Division and chaired by the MSE Division Commander; the SMEs decided that it would be appropriate to hold the oral board in the Diesel Inspection Cell rather than the TE as they represent the coastal experts on the Diesel Inspection Programme. However, the oral exam question bank and preparation of questions prior to each oral board will be managed by the TE on each coast.

10. The board agreed that the US Navy Diesel Engine Inspector Handout Part II would be an essential asset for the conduct of this training

11. Following a lengthy discussion regarding the previous selection methods for the DI course, SME were adamant that a screening process be implemented to ensure that the training was not being received by personnel who would not be employed in the Diesel Inspection Cells throughout their career. SMEs agreed that a course prerequisite be implemented that required nominations for the course to be pre-screened by the applicable coastal Chief Diesel Inspector who would forward their recommendations to the Career Manager.

<u>Lessons Learned:</u>

12. It would be beneficial if the following documents were forwarded to the SMEs with the Convening Order so that they may become familiar with their content before the QSPWB is convened:

a. All applicable JDs; and

b. The CFNES QSPWB Aide Memoire.

<u>Guidance to Instructional Designers:</u>

13. The CFNES MSE Diesel Training Senior Instructor will develop the PC scenarios for PO 002 Troubleshoot Diesel Engines from the table created during the QSPWB with assistance from the CDIs on each coast.

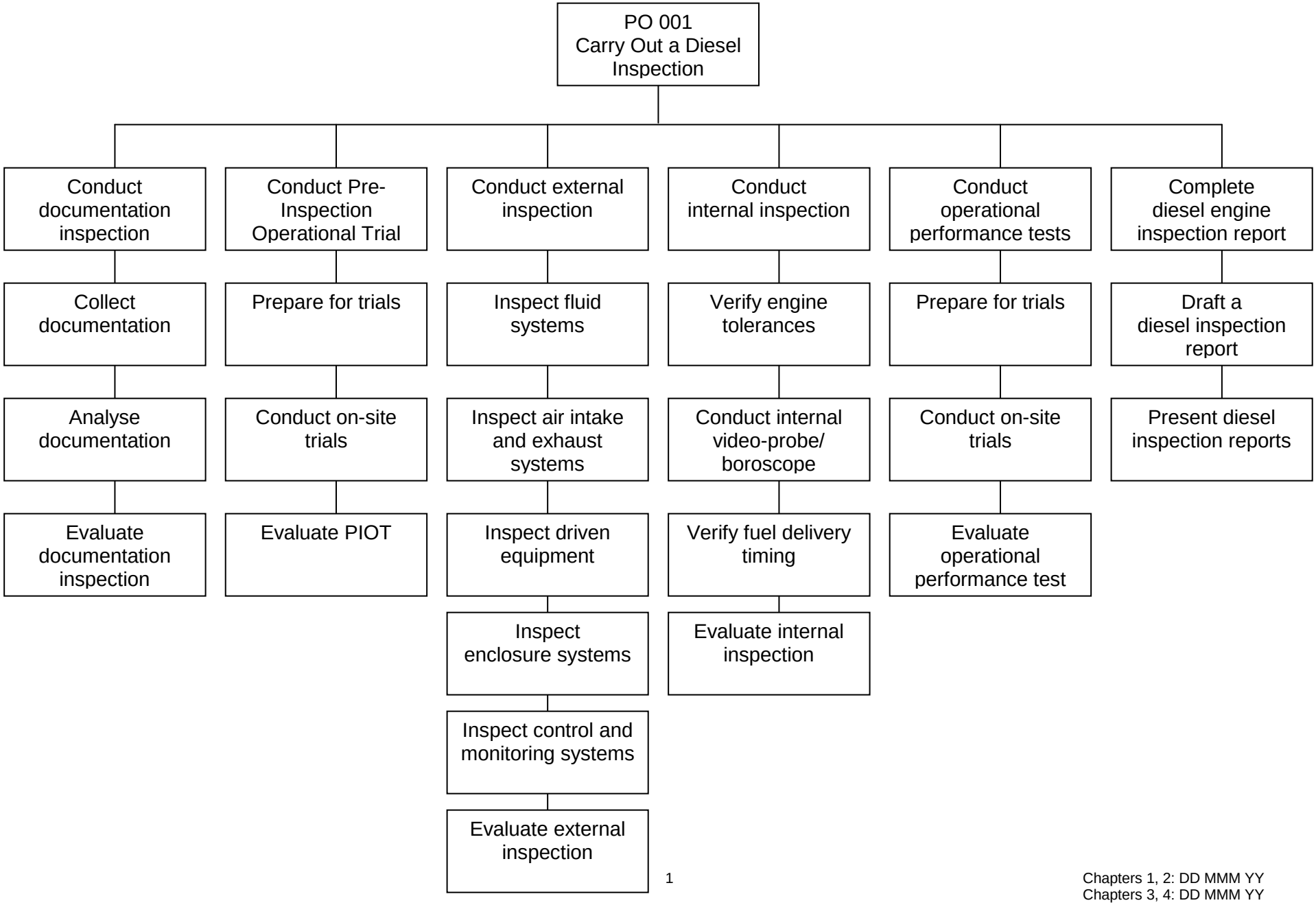
14. The CFNES MSE Diesel Training Senior Instructor will complete the Critical Training Resources table in Chapter One of this QSP, including the training days each resource is required to conduct the course.

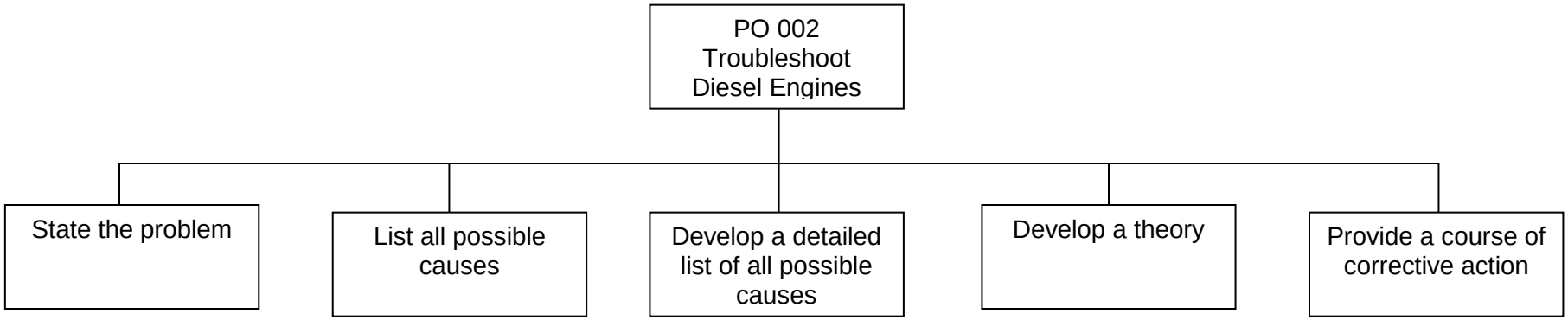
Recommendations / Follow-up Action:

ITEM	ACTION ITEM	OPI(s)
1	Obtain a National Qualification Code for the Diesel Inspection course	DMTE
2	The Training Authority (TA) to order the US Navy Reference book; S9233-CJ-HBK-020 US Navy Diesel Engine Inspector Handout Part II.	DMTE
3	Amend JDs (See ROP Para 5 for modified JDs)	D MAR PERS
4	Amend CFTO C-30-010-551/AM-001 MARINE DIESEL INSPECTION PROGRAMME (See ROP Para 6)	DMTE
5	Task FT0557: Employ electrical / electronic / digital precision measuring test equipment; remove due to duplication in MT2383, MT2384 and MT2385.	D MAR PERS
6	Task FT0818: Inspect diesel engines; remove due to duplication in BT0294.	D MAR PERS
7	Task HT1537: Test alarm; remove due to duplication in DT0566.	D MAR PERS
8	Amend task DT0509; Evaluate trials results to read “Analyse trials results.”	D MAR PERS
9	Amend task FT0821; Test and adjust diesel injectors, to read “Verify fuel delivery timing.”	D MAR PERS
10	Amend task FT0827; Measure engine clearances, to read “Verify engine tolerances.”	D MAR PERS

ITEM	ACTION ITEM	OPI(s)
11	Addition of the following tasks in the JBS: QSPWB0001 – Conduct Documentation Inspection. QSPWB0002 - Conduct Pre-inspection operational trial. QSPWB0003 - Conduct External Inspection. QSPWB0004 – Conduct Internal Inspection. QSPWB0005 - Conduct Operational Performance test. QSPWB0006 – Evaluate Documentation. QSPWB0007 – Evaluate PIOT. QSPWB0008 – Collect documentation. QSPWB0009 – Analyse documentation. QSPWB0010 – Inspect fluid systems. QSPWB0011 – Inspect air intake and exhaust systems. QSPWB0012 - Inspect driven equipment. QSPWB0013 – Inspect control and monitoring systems. QSPWB0014 – Evaluate external inspection. QSPWB0015 – Conduct internal video-probe / boroscope inspection. QSPWB0016 - Evaluate internal inspection. QSPWB0017 – Evaluate operational performance tests.	D MAR PERS

ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR




PO 002
Troubleshoot
Diesel Engines
State the problem
List all possible causes
Develop a detailed list of all possible causes
Develop a theory
Provide a course of corrective action

ANNEX E: ON-JOB PERFORMANCE RECORD

(CERTIFICATE OF QUALIFICATION)

Name and Initials	Rank	SN

Date Training Started	Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR hereinafter referred to as the master copy, will be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training establishment shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress.

2. As the trainee completes each Performance Objective (PO) to the required standard, the Training Assessor will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

4. Upon completion of two supervised diesel inspections the following action will be taken:

   a. The trainees' Supervisor will convene an Oral Board.

   b. The Oral Board:

      (1) Board composition shall include, as a minimum, the SSO MS or representative (East Coast), MSEO or representative from FMF Engineering (West Coast), Chief Diesel Inspector (CDI) and the trainees' supervisor;

      (2) Board questions shall be supplied from CFNES Standards Section. A minimum of 6 questions to be provided to the respective CDI no later than 24 hours prior to board date. The following is a breakdown of the type of questions to be asked;

         (a) Diesel Inspection Report (DIR) - The first two questions shall pertain to the trainees' two completed DIR's and shall be supplied by the respective coastal CDIs. One question shall require the trainee to sketch and describe a system related to one of the candidates supervised diesel inspections;

(b) Diesel Technical Knowledge – Two questions shall be related to diesel technical knowledge. CFNES will supply the two required questions, plus one supplemental question;

(c) Administrative Questions – One question will be asked pertaining to the diesel administrative process of the Diesel Inspection Programme. CFNES will supply the one required question, plus one supplemental question;

(d) Current Fleet Diesel Issues – One question will be asked pertaining to current diesel technical issues in the fleet and shall be supplied by the respective coastal CDIs; and

(e) A maximum of two supplementary questions may be asked at the discretion of the board members.

(3) On completion of the oral board the completed OJPR (ANNEX E) shall be forwarded to the applicable coastal Quality Control Section; and

(4) Upon successful completion of the OJPR, the applicable coastal Quality Control Section shall award the qualification and forward the trainees' results by message to the parent unit and D-MIL C.

c. Administrative (all candidates):

(1) The Chief Diesel Inspector shall complete section 6 of the OJPR Certificate of Qualification;

(2) The individual's Qualification Record Sheet (QRS) shall be completed as described in A-PM-245-001/FP-001, Military HR Records Procedures, and distributed to the respective MARLANT/MARPAC Headquarters for action; and

(3) The OJPR shall be retained on the individual's UER until update action is reflected on the member's Personal Record Resume (PRR).

TRAINING COMPLETION CERTIFICATE

5. Date training completed: _____

_____

Chief Diesel Inspector Date

CERTIFICATION OF QUALIFICATION

6. I certify that _____ has completed training to the standards stated in the QSP.

_____

Divisional Officer Date

PO 001 Carry Out a Diesel Inspection		
EO 001.01 to EO 001.06 Carry Out a Diesel Inspection		
❖ Refer to the standard statements for EO 001.01 through to EO 001.06 for requirements		
Part I **Conduct First Supervised Diesel Inspection**		
	Remarks	DATE / SIGNATURE
a. **EO 001.01** Phase One – Conducting Documentation Inspection		
o Collecting documentation		
o Analysing documentation		
o Evaluating documentation		
b. **EO 001.02** Phase Two – Conducting PIOT		
o Preparing for trials		
o Conducting on-site trials		
o Evaluating PIOT		
c. **EO 001.03** Phase Three – Conducting External Inspection		
o Inspect Fluid Systems		
o Inspect Air Intake and Exhaust System		
o Inspect Engine Driven Equipment		
o Inspect Engine Enclosures		
o Inspect Control and monitoring systems		
o Evaluating external inspection		
d. **EO 001.04** Phase Four – Conducting Internal Inspection		
o Verifying engine tolerances		
o Conducting internal video-probe / boroscope inspection		
o Verifying fuel delivery timing		
o Evaluating Internal Inspection		
e. **EO 001.05** Phase Five – Conducting Operational Performance Test		
o Preparing for trials		
o Conducting on-site trials		

o Evaluating operational performance tests		
f. **EO 001.06** Phase Six – Completing Draft and Submit Diesel Inspection Report		

PO 001 Carry Out a Diesel Inspection		
EO 001.01 to EO 001.06 Carry Out a Diesel Inspection		
❖ Refer to the standard statements for EO 001.01 through to EO 001.06 for requirements		
Part II **Conduct Second Supervised Diesel Inspection**		
	Remarks	DATE / SIGNATURE
a. **EO 001.01** Phase One – Conducting Documentation Inspection		
o Collecting documentation		
o Analysing documentation		
o Evaluating documentation		
b. **EO 001.02** Phase Two – Conducting PIOT		
o Preparing for trials		
o Conducting on-site trials		
o Evaluating PIOT		
c. **EO 001.03** Phase Three – Conducting External Inspection		
o Inspect Fluid Systems		
o Inspect Air Intake and Exhaust System		
o Inspect Engine Driven Equipment		
o Inspect Engine Enclosures		
o Inspect Control and monitoring systems		
o Evaluating external inspection		
d. **EO 001.04** Phase Four – Conducting Internal Inspection		
o Verifying engine tolerances		
o Conducting internal video-probe / boroscope inspection		
o Verifying fuel delivery timing		
o Evaluating Internal Inspection		
e. **EO 001.05** Phase Five – Conducting Operational Performance Test		
o Preparing for trials		

o Conducting on-site trials		
o Evaluating operational performance tests		
f. **EO 001.06** Phase Six – Completing Draft and Submit Diesel Inspection Report		

PO 001 Carry Out a Diesel Inspection	
EO 001.01 to EO 001.06 Carry Out a Diesel Inspection	
❖ Refer to the standard statements for EO 001.01 through to EO 001.06 for requirements	
Part III **Oral Board**	
Remarks	DATE / SIGNATURE
To be signed by CDI.	

ANNEX F: ACRONYMS / ABBREVIATIONS

ACRONYM / ABBREVIATION	TERM/DEFINITION
DDFP	Diesel Driven Fire Pump
DG	Diesel Generator
EDEA	Electronic Diesel Engine Analyzer
EHM	Equipment Health Monitoring
MASIS	Material Acquisition & Support Information System
MDEL	Marine Diesel Engine Log
CERA	Chief Engineering Room Assistant
CDI	Chief Diesel Inspector
PPE	Personal Protective Equipment
OEM	Operation Engineering Maintenance
OJE	On Job Experience
OJPR	On Job Performance Record
DG LOG BOOK	Diesel Generator
CFTO	Canadian Forces Technical Order
EHM RECORD	Equipment Helt Monitoring
DIP	Diesel Inspection Programme
PIOT	Pre-Inspection Operational Trial
DI	Diesel Inspector
PM/CM	Preventive Maintenance / Corrective Maintenance
CMIS-S/DRMIS	Career Management Information System / Defence Resource Management Information System
OCCAP	Oil and Coolant Condition Analysis Program
FMFCS	Fleet Maintenance Facility Cape Scott
SS	Ship Staff

ACRONYM / ABBREVIATION	TERM/DEFINITION
PO	Performance Objective
PC	Performance check
EO	Enabling Objective
EC	Enabling check

ANNEX G: SUGGESTED SEQUENCING AND TIMINGS

PO/EO	OBJECTIVE	EO	EC	PC	PO

PO/EO	OBJECTIVE	EO	EC	EO	PC	TOTAL

[Or insert appropriate sample schedule] .

ANNEX H: TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

ANNEX I: COSTING INFORMATION

Include information about cost in whichever format is applicable.

This annex is optional (**Not** to be deleted).

ANNEX J: COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments is to be provided here.

DI CON ORD AMEND 002.ocr

DI exec summary

ANNEX K: EVALUATION MATRIX

PC / EC	SCOPE	PURPOSE	HOW	WHEN	TIME	RESOURCES	LIMITATIONS	IMPACT
EC 001.03		To evaluate candidates ability to carry out phases 1 to 3 of a DI.	Written test Multiple choice questions	At completion of EO 001.03	50 Min Including review by instructor	Nil	Nil	Trainees will require review with instructor
EC 001.06		To evaluate candidates ability to carry out phases 4 to 6 of a DI.	Written test Multiple choice questions	At completion of EO 001.06	50 Min Including review by instructor	Nil	Nil	Trainees will require review with instructor
PC 001		To evaluate candidates ability to carry out a Diesel Inspection	Written test Multiple choice questions Long / short answered questions.	At completion of PO 001	100 Min	Nil	Nil	Trainee will require remedial training followed by a retest. Subsequent failure will require a TRB
PC 002		To evaluate candidates ability to trouble shoot diesel engines	Practical test Using two scenarios per student of progressive difficulty with one fault each.	At completion of PO 002	50 Min per scenario	Applicable CFTO, Tools and test equipment, Small and large test cell.	Limited assistance as determined by instructor as required	Trainee will require remedial training followed by a retest. Subsequent failure will require a TRB

ANNEX L: TROUBLESHOOTING PROCESS CHECKLIST

TROUBLESHOOTING PROCESS CHECKLIST		
Checklist Item	**SAT/ UNSAT**	**Comments**
State the problem • Precisely state the problem and not the symptoms. (The problem statement must be clearly defined)		
List all possible causes • List major system • List components • List reasons		
Develop a detailed list of all possible causes. • Engine familiarization-design, construction, installation and application. • Add or eliminate possible causes. • Collect applicable information. o History o Logs o Performance Tests o Running Hours o Number of Failures o Problem Discovery Method o What happened this time o Previous Maintenance o Recent Repairs o Corrective measures since failure		
Develop a theory • Develop a plan of action, simplest to hardest • Begin the process of elimination • Add additional causes if discovered during inspection • Use process of elimination until actual cause and not just the result is found • Re-evaluate selection • Review list to ensure cause is the only logical cause • Ensure selection justifies how problem was caused. Support		

TROUBLESHOOTING PROCESS CHECKLIST		
Checklist Item	**SAT/ UNSAT**	**Comments**
finding with facts and logical reasoning based on OJE and thorough understanding of diesel basics.		
Providing a course of corrective action • Give solid recommendations to eliminate the problem		

Overall Assessment SAT/UNSAT		
Instructor Comments:		
Date	**Name**	**Signature**

Administrative Instructions:

11. The Diesel Engine Troubleshooting assessment Practical Assessment Checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statement from PO 002 – Troubleshoot Diesel Engine.

Guidance for Instructors/Assessors:

12. The following guidance is provided to Instructors/Assessors for administering the PC assessment:

a. Instructors/Assessors will brief the trainees on the following regarding the PC:

(1) Procedures; and

(2) Safety.

b. Instructors/Assessors will conduct the PC by:

(1) Briefing trainee on scenario;

(2) Executing scenario (50 min max);

(3) On conclusion of scenario:

(a) Debrief the trainee being assessed; and

(b) Set up for next assessment.

c. Instructors/Assessors will ensure the following resources are obtained/in place prior to running the PC:

(1) PC checklist;

(2) Fault has been installed into diesel;

(3) Trainer is flashed up;

(4) DTF personnel know what the active scenario is, specifically:

(a) Lead instructor acting as assessor; and

(b) Safety numbers.

d. Instructors/Assessors will determine the standard for achievement based on the Practical Assessment Checklist, with a performance considered Satisfactory only after all sections are considered satisfactory.

e. Post assessment action will include the Instructors/Assessors:

(1) Informing the trainee being assessed that they were successful or unsuccessful in the PC; and

(2) Provide areas for improvement for the trainee.

f. In determining the level of assistance Instructors/Assessors may provide trainees, the following should be considered:

(1) Try to ensure the trainee is given an opportunity for the scenario to progress to its conclusion;

(2) If the trainee experiences difficulty, provide minimal assistance to maintain progression of the PC; and

(3) Safety shall be observed throughout. If a safety infraction is committed or about to be committed the scenario should be either:

(a) Paused and corrective action taken before continuing; or

(b) Halted if the infraction is serious in nature.

ANNEX M: TROUBLESHOOTING GUIDE

1. **State the problem as described.**

2. **List all possible causes;**

    a. List major system, components, or reasons.

3. **Develop a detail list of all possible causes;**

    a. Familiarize yourself with the engine design, construction, installation and application;

    b. Add or eliminate possible causes resulting from step a;

    c. Obtain from reliable sources / or numerous sources;

        i. Engine; background, history, logs, performance tests, basic information (Hrs, since failure) and number of failures.

        ii. Answer the following questions;

            1. How was the problem discovered?

            2. What happened at this time?

            3. What maintenance has been performed, routine or preventive?

            4. What repair etc. has been done recently?

            5. What have they done since the failure to correct the problem/failure?

    d. Based upon all the information obtained above, develop a detailed list of all possible causes/ reasons for the reported problem/failure. Re-Define the problem if required.

NOTE: DO NOT ELIMINATE CAUSES SOLELY BECAUSE THE SHIP HAS TOLD YOU THAT THEY HAVE ALREADY CHECKED/CORRECTED THIS POSSIBLE CAUSE!!

4. **Develop a plan of action**.

    a. List simplest / easiest causes first.

    b. Begin the process of elimination by:

        i. Inspecting, listing, observing, analyzing, checking, and evaluating each item on your list of causes.

c. Add additional causes for the problem if discovered during your inspection/investigation of engine, systems and components and etc.

d. Continue down the list until the actual cause and not just a result is found.

e. Review your list to guarantee that your cause is the only logical, possible and reasonable cause.

f. Re-evaluate your selection. Make sure that in your mind, that you are satisfied with your choice.

5. **Develop a theory** (if your cause is unusual or not obvious) on how your selection caused the problem (failure, explosion etc.). Have sufficient justification to prove or explain your theory. Support your finding/theory with the facts and logical reasoning (Common sense, technical experience and thorough understanding of diesel engine basics.

6. **Provide a course of corrective action and solid recommendations which will eliminate / correct the problem.**

NOTE: THE ABOVE STATEMENT WILL ONLY BE VALID IF YOU DETERMINED THE ACTUAL CAUSE OF THE PROBLEM AND NOT JUST A RESULT, PRODUCT OR CONSEQUENCE OF THE ACTUAL CAUSE.